Computer Networks Spring 2026
Assignment#6 (6A & 6B)

Due Date: Tuesday, 5th May, 2026
Submission Mode & Time: Handwritten solutions to be submitted during the lecture.

Please note the following:

1. No exceptions to the above date and time will be allowed. Inability to submit the assignment by the required time will result in zero marks.
2. To ensure self-completion of assignments and discourage plagiarism, the instructor or the relevant TA may randomly contact you and ask for an explanation of your answers. Where plagiarism and/or cheating is evident, you will be referred to the departmental disciplinary committee. In extreme cases of plagiarism an F may be awarded immediately with further referral to the university disciplinary committee.
3. All solutions must be **hand-written.**
4. **Assignment Solution Submission**: In case of **in person / physical lectures at the campus**, hard copy of the hand-written assignment's solutions will be submitted by **hand** by each student to the Instructor / TA directly during the lecture on the due date.

PART-1

Use the following text for completion of this part of the assignment:
Computer Networking - A Top-Down Approach 8th Edition by Kurose & Ross.

Solve the following problems from the back of **Chapter 6**. Every Question has equal marks i.e.

Problems: **(4*6 = 24 marks) [CLO 3]**
P1, P5, P6, P14, P22, P26

PART - 2

Question: 01 [10 Marks] [CLO 3]

Consider the cyclic redundancy check (CRC) based error-detecting scheme with the generator polynomial $x^3 + x + 1$.

Suppose the message $m_4m_3m_2m_1m_0 = 11000$ is to be transmitted. The check bits c_2, c_1, c_0 are appended at the end of the message by the transmitter using the CRC scheme.

The transmitted bit string is denoted by $m_4m_3m_2m_1m_0c_2c_1c_0$

The task is to find the value of the check bit sequence $c_2c_1c_0$.

Question: 02 [5+5 = 10 Marks] [CLO 3]

In a Slotted ALOHA network with N active nodes, the maximum efficiency is approximately 37% (1/e).

a. Why does **CSMA/CD** achieve significantly higher efficiency than Slotted ALOHA?
b. What specific characteristic of the physical medium (propagation delay) limits the efficiency of CSMA/CD?

Question: 03 [6 Marks] [CLO 3]

A student opens their laptop in a coffee shop and types www.google.com into their browser. The laptop just connected to the WiFi and has no cached data. Order the following events chronologically (1-6) and briefly explain the dependency (why X must happen before Y).

A. DNS Query is sent to the DNS Server to resolve www.google.com.
B. Laptop sends a DHCP Discovery message to get an IP address.
C. Laptop sends a TCP SYN segment to the Google Web Server.
D. Laptop sends an ARP Request to find the MAC address of the Default Gateway.
E. Laptop receives a DHCP Offer/Ack with its IP, Gateway IP, and DNS IP.
F. Laptop sends an HTTP GET request.

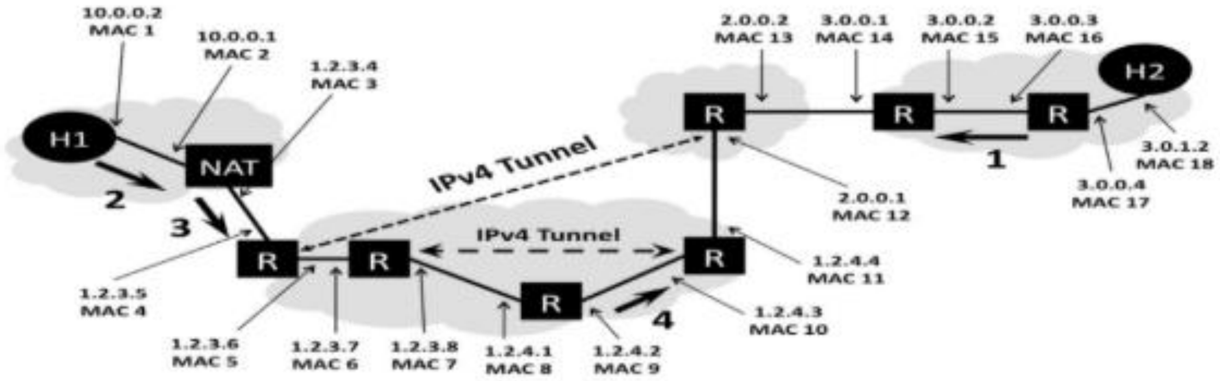
Question: 04 [5+5 = 10 Marks] [CLO 3]

Suppose four active nodes—nodes A, B, C and D—are competing for access to a channel using slotted ALOHA. Assume each node has an infinite number of packets to send. Each node attempts to transmit in each slot with probability 0.2. The first slot is numbered slot 1, the second slot is numbered slot 2, and so on.

a. What is the probability that node A succeeds in slot 1?
b. What is the probability that node A succeeds for the first time in slot 4?

Question: 05 [1x10 = 10 Marks] [CLO 3]

The figure below shows a network topology. The LAN on the left uses a NAT to connect to the Internet and includes a client host H1. The LAN on the right includes a webserver H2. Packets between the two endpoints are routed along the path shown by a heavy dark line, which includes two IPv4 tunnels. All packets which traverse the path use both tunnels. The various network interfaces have IP and MAC addresses as shown:



H1 has established an HTTP session with web server H2 and data packets are flowing between the two machines. You have to fill in the missing entries in the tables below, i.e. the source and destination address for the corresponding network and datalink layers for packets 2, 3, and 4 (these packets are all traveling from the client H1 to the server H2, as marked on the figure with heavy black arrows and numbers).

Packet 1		
Header	**Source**	**Destination**
Ethernet		
IP Address		

Packet 2		
Header	**Source**	**Destination**
Ethernet		
IP Address		

Packet 3		
Header	**Source**	**Destination**
Ethernet		
IP Address		

Packet 4		
Header	**Source**	**Destination**
Ethernet		
IP Address		